INTELLIGENT HOUSE PRICE ESTIMATION SYSTEM

Database Requirement Analysis

Overview

The database for the Intelligent House Price Estimation System is designed to store, manage, and retrieve all data related to house properties, user interactions, ML-based price predictions, and transaction records. The following tables form the core structure of the system's database.

Table List Summary

#	Table Name	Description
1	users	Stores user and admin login details
2	houses	Stores core house property details
3	locations	Stores geographic and area information
4	price_estimations	Stores AI-predicted price results
5	features	Stores specific house features for prediction
6	amenities	Stores facilities available in/near the house
7	feedback	Stores user reviews and ratings
8	transaction_history	Stores past house sale records for ML training

Detailed Table Descriptions

Table 1: users

This table stores all registered users and administrators of the system. It manages authentication, role-based access, and user identity.

Column	Data Type	Constraint	Description
user_id	INT	PRIMARY KEY	Unique identifier for each user
username	VARCHAR(100)	NOT NULL	User's full name
email	VARCHAR(150)	UNIQUE, NOT NULL	Email address for login
password	VARCHAR(255)	NOT NULL	Encrypted password
role	VARCHAR(20)	DEFAULT "user"	Role: Admin or User
created_at	DATE	DEFAULT NOW()	Account registration date

Table 2: houses

This table stores the core property details of each house listed in the system. It serves as the central entity linked to most other tables.

Column	Data Type	Constraint	Description
house_id	INT	PRIMARY KEY	Unique identifier for each house
location_id	INT	FOREIGN KEY	References locations table
bhk	INT	NOT NULL	Number of bedrooms/hall/kitchen
square_feet	FLOAT	NOT NULL	Total area in square feet
floor	INT	NOT NULL	Floor number of the property

age	INT	NOT NULL	Age of the building in years
price	FLOAT	NOT NULL	Actual or listed price of house

Table 3: locations

This table stores geographic and area-based data for each property. Location is a major factor in house price estimation.

Column	Data Type	Constraint	Description
location_id	INT	PRIMARY KEY	Unique identifier for each location
city	VARCHAR(100)	NOT NULL	Name of the city
locality	VARCHAR(150)	NOT NULL	Area or neighborhood name
pincode	VARCHAR(10)	NOT NULL	Postal/ZIP code
state	VARCHAR(100)	NOT NULL	State or province name

Table 4: price_estimations

This table records each price prediction made by the ML model. It links users and houses to track who requested which estimation and when.

Column	Data Type	Constraint	Description
estimation_id	INT	PRIMARY KEY	Unique ID for each estimation
user_id	INT	FOREIGN KEY	References users table
house_id	INT	FOREIGN KEY	References houses table
predicted_price	FLOAT	NOT NULL	Price predicted by ML model
estimation_date	DATE	DEFAULT NOW()	Date the estimation was made

Table 5: features

This table captures specific house features that influence price prediction such as furnishing, balcony availability, parking, and bathrooms.

Column	Data Type	Constraint	Description
feature_id	INT	PRIMARY KEY	Unique ID for each feature record
house_id	INT	FOREIGN KEY	References houses table
balcony	BOOLEAN	DEFAULT FALSE	Whether balcony is available
furnished	VARCHAR(20)	NOT NULL	Furnished / Semi / Unfurnished
parking	BOOLEAN	DEFAULT FALSE	Whether parking is available
bathrooms	INT	NOT NULL	Total number of bathrooms

Table 6: amenities

This table stores information about extra facilities available in or near the house. Amenities significantly affect property valuation.

Column	Data Type	Constraint	Description
amenity_id	INT	PRIMARY KEY	Unique ID for each amenity record
house_id	INT	FOREIGN KEY	References houses table
gym	BOOLEAN	DEFAULT FALSE	Gym facility available
swimming_pool	BOOLEAN	DEFAULT FALSE	Swimming pool available

school_nearby	BOOLEAN	DEFAULT FALSE	School in nearby vicinity
hospital_nearby	BOOLEAN	DEFAULT FALSE	Hospital in nearby vicinity

Table 7: feedback

This table collects user reviews and ratings about the system performance and accuracy of price estimations.

Column	Data Type	Constraint	Description
feedback_id	INT	PRIMARY KEY	Unique ID for each feedback
user_id	INT	FOREIGN KEY	References users table
rating	INT	CHECK (1-5)	Rating score out of 5
comments	TEXT	NULLABLE	User written comments
feedback_date	DATE	DEFAULT NOW()	Date feedback was submitted

Table 8: transaction_history

This table stores historical house sale data used for training and improving the ML price estimation model.

Column	Data Type	Constraint	Description
transaction_id	INT	PRIMARY KEY	Unique ID for each transaction
house_id	INT	FOREIGN KEY	References houses table
sold_price	FLOAT	NOT NULL	Final price at which house was sold
sale_date	DATE	NOT NULL	Date the transaction occurred
buyer_id	INT	FOREIGN KEY	References users table (buyer)

Table Relationships Summary

Table	Related To	Relationship Type
users	price_estimations, feedback, transaction_history	One-to-Many
houses	features, amenities, price_estimations, transaction_history	One-to-Many
locations	houses	One-to-Many
price_estimations	users, houses	Many-to-One
transaction_history	houses, users	Many-to-One